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I am a Research Specialist working at King Abdullah University of Science and Technology (KAUST), currently on leave from my position as an Assistant Professor in Statistics at Universiti Brunei Darussalam (UBD). My expertise lie in statistical methodology and computation, with applications towards the natural and social sciences.

I am passionate about bringing data and technology to the forefront of our daily interaction with the world. Using the R programming language, I build statistical models for data analysis in order to gain insights, make predictions, and aid decision-making. I am frequently engaged in cross-disciplinary research collaborations and statistical consultations.

Research Interests

Latent variable models

Composite likelihood methods

Spatio-temporal models

Bayesian estimation

Integrated Nested Laplace
Approximation (INLA)

Hedonic pricing models

Education

2014 — 2018

PhD in Statistics, London School of Economics and Political Science

Supervisors: Wicher Bergsma¹ and Irini Moustaki²

*Thesis: Regression modelling using priors depending on Fisher information covariance kernels (I-priors)*³

2013 — 2014

MSc in Statistics (distinction), London School of Economics and Political Science

2006 — 2010

BSc & Master in Mathematics, Operational Research, Statistics and Economics (first class honours), University of Warwick

Supervisor: David Firth⁴

Dissertation: The analysis of paired comparison data using Bradley-Terry models with applications to football data

Awards & honours

2023

Teaching Excellence Award in the Sciences, Universiti Brunei Darussalam

2020

Arnold Zellner Thesis Award in Econometrics and Statistics (honourable mention)⁵, American Statistical Association (Business and Economic Statistics Section)

2013 — 2018

In-Service Training Scheme Scholarship, Public Service Commission, Prime Minister's Office, Brunei

2006 — 2010

Supreme Commander of The Royal Brunei Armed Forces Scholarship⁶, Ministry of Defence, Brunei

Employment

- 2025 — present **Research Specialist**
King Abdullah University of Science and Technology⁷
I conduct research on efficient Bayesian computational methods for latent variable modeling within the BAYESCOMP group led by Prof. Håvard Rue.
- 2024 — 2027 **Visiting Fellow**
Department of Statistics, London School of Economics and Political Science⁸
The School's Visiting Appointments Scheme is a device to help Departments, Institutes and Centres invite staff who are either practitioners or academics in other institutions to conduct research and other activities.
- 2019 — present **Assistant Professor in Statistics**
Mathematical Sciences, Faculty of Science, *Universiti Brunei Darussalam*⁹
[Leave of absence from August 2025]

I teach SM-1402 Basic Statistics, SM-2302 Software for Mathematicians, SM-4331 Advanced Statistics, SM-4335 Advanced Probability, and CU-0304 Sciences for University Study. I also supervise mathematics students' final year projects. In addition to my teaching duties, I am the current Postgraduate Coordinator (Mathematics). I previously held the post of Discovery Year¹⁰ advisor (2019-2021).
- 2010 — 2014
2018 — 2019 **Graduate Teaching Assistant**
Department of Statistics, London School of Economics and Political Science¹¹
I held various teaching roles ranging from secondary teaching to marking for masters (ST416 Multilevel Modelling, ST421 Development in Statistical Methods) and undergraduate level courses (ST312 Applied Statistics Project) at the LSE. In addition, I also held assistant invigilator and examiner roles for both the LSE and the University of London International Examinations respectively.
- 2010 — 2019 **Research Officer**
Centre for Science and Technology, Research and Development (CSTRAD), Ministry of Defence, Brunei¹²
I provided analytical insight into matters arising from strategic procurement activities of the Royal Brunei Armed Forces (RBAF). This entailed market research, cost-benefit analyses, and writing technical reports. I was a core team member for the procurement and delivery of 12 Sikorsky S-70i helicopters¹³ (US\$325M), and also the focal point for the development of tender evaluation criteria using the Analytic Hierarchy Process.

Service

- 2024 — 2026 **Chair**
Brunei R User Group¹⁴
- 2020 — present **Ad-hoc Reviewer**
Annals of Applied Statistics¹⁵; Journal of the Royal Statistical Society: Series C¹⁶
- 2020 — present **Review Editor**
Frontiers in Applied Mathematics and Statistics¹⁷
- 2020 **Guest Lecturer**
Universiti Teknologi Brunei¹⁸

- 2014 — 2015 **MSc Programme Facebook Administrator**
Department of Statistics, London School of Economics and Political Science
Acted as student liaison for incoming cohort of MSc Statistics 2015/16.
- 2013 — 2015 **Auditor**
Brunei Down's Syndrome Association (ABLE)¹⁹

Consultancy

Grants and funding

- 2021 — 2024 **Extension of I-prior methodology to nominal response models** (~BND 7,000), Faculty of Science Research Grant, *Universiti Brunei Darussalam*.
Role: *Principal Investigator*
- 2021 — 2023 **Examining spillover effects in regional trading blocs** (~BND 17,000), School of Business and Economics Research Grant, *Universiti Brunei Darussalam*.
Role: *Member*
- 2021 — 2024 **Mathematical Modeling of Genetic Assimilation** (~BND 8,000), Faculty of Science Research Grant, *Universiti Brunei Darussalam*.
Role: *Co-Principal Investigator*
- 2023 — 2024 **Predictive potential of graphical descriptors for QSPR/QSAR model learning of compounds** (~BND 15,700), Faculty of Science Research Grant, *Universiti Brunei Darussalam*.
Role: *Co-Principal Investigator*
- 2024 **R User Groups (RUGS) Program Grants** (USD 500), R Consortium²⁰

Research output

Articles (peer-reviewed)

- 2025 A18 **Conditionally Autoregressive Models for House Price Data: Insights from a Comparative Simulation Study**
I. Puteri Kinasih, H. Jamil, E. Leong
Journal of Statistical Theory and Applications. <https://doi.org/10.1007/s44199-025-00122-1>
- A17 **Leveraging sparse Gaussian processes for property price modelling and sustainable urban planning**
H. Jamil, F. Usop, H. M. Ramli
In: S. A. Abdul Karim & A. Baharum (Eds.), *Intelligent Systems of Computing and Informatics in Sustainable Urban Development*. Taylor and Francis/CRC Press.
<https://doi.org/10.1201/9781003518396-2>
- A16 **A spatio-temporal analysis of house prices in Brunei Darussalam**
H. Jamil
Quality & Quantity. <https://doi.org/10.1007/s11135-025-02164-0>
- A15 **From archives to AI: Residential property data across three decades in Brunei Darussalam**
H. Jamil, A. Barizah Noorosmawie, H. Waezz Rabu, L. Abdul Razak
Data in Brief, 60, 111505. <https://doi.org/10.1016/j.dib.2025.111505>

- 2024 A14 **An Exploratory Investigation on Foreign Language Anxiety among English for Islamic Studies Learners in Brunei Darussalam**
N. Noorashid, *H. Jamil*
Electronic Journal of Foreign Language Teaching, 21(2) 211-234. <https://doi.org/10.56040/nnhj2123>
- A13 **Pairwise likelihood estimation and limited-information goodness-of-fit test statistics for binary factor analysis models under complex survey sampling**
H. Jamil, I. Moustaki, C. Skinner
British Journal of Mathematical and Statistical Psychology, 78(1), 258–285. <https://doi.org/10.1111/bmsp.12358>
- A12 **An Optimization Problem for Computing Predictive Potential of General Sum/Product-Connectivity Topological Indices of Physicochemical Properties of Benzenoid Hydrocarbons**
S. Hayat, A. Arfan, A. Khan, *H. Jamil*, M. J. F. Alenazi.
Axioms 13(6). <https://doi.org/10.3390/axioms13060342>
- 2023 A11 **Importance of Baseline Assessments: Monitoring of Sungai Brunei River's Water Quality**
O. Onifade, N. Shamsuddin, *H. Jamil*, D. T. C. Lai, S. Gödeke
H2Open Journal, 6(4) 518–534. <https://doi.org/10.2166/h2oj.2023.168>
- A10 **Extremal hyper-Zagreb index of trees of given segments with applications to regression modeling in QSPR studies**
S. Hayat, M. Adil Khan, A. Khan, *H. Jamil*, M. Yasir Hayat Malik
Alexandria Engineering Journal, 80(1) pp.259–268
- A09 **Statistical significance of valency-based topological descriptors for correlating thermodynamic properties of benzenoid hydrocarbons with applications**
S. Hayat, N. Suhaili, *H. Jamil*
Computational and Theoretical Chemistry, 114259. <https://doi.org/10.1016/j.comptc.2023.114259>
- A08 **The domination number of the king's graph**
M. Arshad, S. Hayat, *H. Jamil*
Computational and Applied Mathematics, 42(251). <https://doi.org/10.1007/s40314-023-02386-8>
- 2022 A07 **Factors Affecting Readmission to a Drug Rehabilitation Centre**
F. Jali, E. Leong, *H. Jamil*
Journal of Population and Social Studies, 30, 591-601.
- A06 **Iron and Manganese Mobilisation due to Dam Height Increase for a Tropical Reservoir in South East Asia**
S. Gödeke, *H. Jamil*, M. Schirmer, A. Bretzler, N. Shamsuddin, N. H. Mansor
Environmental Monitoring and Assessment, 194(5), 358.
- A05 **Pairwise likelihood estimation for confirmatory factor analysis models with categorical variables and data that are missing at random**
M. Katsikatsou, I. Moustaki, *H. Jamil*
British Journal of Mathematical and Statistical Psychology, 75(2) 23-45.
-  **Wiley Top Downloaded Article 2021**

- 2021 A04 **Advocating Blended Learning for University Undergraduate Level Mathematical Instruction Beyond Covid-19**
H. Jamil, H. M. Ramli, E. Leong
 In: Engineering and Sciences Teaching and Learning Activities: New Systems Throughout COVID-19 Pandemics, pp. 33-45. Ed. by Samsul Ariffin Abdul Karim and Saiful Azmi Hussain. Studies in Systems, Decision and Control, vol 381. Cham: Springer. ISBN: 978-3-030-79613-6.
- A03 **Latent Class Analysis: Insights about design and analysis of schistosomiasis diagnostic studies**
A. Koukounari, H. Jamil, E. Erosheva, C. Shiff, I. Moustaki
 PLOS Neglected Tropical Diseases, 15(2), e0009042.
- 2020 A02 **Bayesian Variable Selection for Linear Models Using I-Priors**
H. Jamil, W. Bergsma
 In: Theoretical, Modelling and Numerical Simulations Toward Industry 4.0, pp. 107–132. Ed. by Samsul Ariffin Abdul Karim. Studies in Systems, Decision and Control, vol 319. Singapore: Springer. ISBN: 978-981-15-8986-7.
- 2018 A01 **Regression modelling using priors depending on Fisher information covariance kernels (I-priors)**
H. Jamil
 PhD Thesis. London School of Economics and Political Science.
 **Arnold Zellner award, honourable mention (top 3)**

Working papers (preprints and unpublished reports)

- 2025 W10 **Bias-Reduced Estimation of Structural Equation Models**
H. Jamil, Y. Rossel, O. Kemp, I. Kosmidis
 arXiv:2509.25419 [stat.ME]
- W09 **A Bayesian factor analytic approach for handling missing data in spatial regressions models**
I. Puteri Kinasih, H. Jamil, E. Leong
 10.13140/RG.2.2.30776.61448
- 2023 W08 **Additive interaction modelling using I-priors**
W. Bergsma, H. Jamil
 arXiv:2007.15766 [math.ST]
- W07 **Patterns of metabolic thermal independence in marine gastropods: A meta-analytical assessment**
M. Akram Ullah, D. Marshall, H. Jamil
 Manuscript in preparation
- 2021 W06 **I think I understand: Investigating misconceptions regarding hypothesis test concepts among university students**
A. Zaini, H. Jamil, E. Leong
 EdArXiv. <https://doi.org/10.35542/osf.io/dj6x9>
- 2020 W05 **Regression modelling with I-priors: With applications to functional, multilevel and longitudinal data**
W. Bergsma, H. Jamil
 arXiv:2007.15766 [math, stat].
- W04 **Investigating the effect of load carriage on soldiers' performances using Bayesian structural equation models**
H. Jamil, A. Salleh, L. Thieng Chan

2019	W03	iprior: An R Package for Regression Modelling using I-priors <i>H. Jamil, W. Bergsma</i> arXiv:1912.01376 [stat].
2017	W02	Discussion contribution <i>W. Bergsma, H. Jamil</i> TI Cunnings, RJ Samworth (2017). Random-projection ensemble classification. JRSSB, 79.4, pp. 959-1035.
2010	W01	Analysis of paired comparison data using Bradley-Terry models with applications to football data <i>H. Jamil</i> Masters Thesis. University of Warwick.

Software

2025	S9	R/plFA: Stochastic Pairwise Likelihood Estimation of Factor Analysis Models <i>G. Alfonzetti, H. Jamil</i> release no releases or repo not found
2024	S6	R/bruneimap: Maps and Spatial Data of Brunei <i>H. Jamil</i> release v0.3.1
	S7	R/birt: Bias-Reduced IRT Models <i>I. Kosmidis, H. Jamil</i> release no releases or repo not found
	S8	R/brlavaan: Bias-Reduced Latent Variable Analysis <i>H. Jamil, Y. Rossel, O. Kemp, I. Kosmidis</i> release no releases or repo not found
2023	S5	R/lavaan.bingof: Binary Goodness-of-Fit Tests for Factor Analysis Models <i>H. Jamil</i> release no releases or repo not found
2021	S4	R/iprior: Regression Modelling using I-priors <i>H. Jamil</i> CRAN 0.7.4 downloads 42K
2020	S3	stat-tables: Open-source statistical tables for use in examinations, tests, and assignments <i>H. Jamil</i> release v0.2 downloads 1.1k
2018	S2	R/ipriorBVS: Bayesian Variable Selection using I-priors <i>H. Jamil</i> release v0.1.1
	S1	R/iprobit: Binary and Multinomial Probit Regression with I-priors <i>H. Jamil</i> release v0.1

Talks

Further information regarding the following events are available at haziqj.ml/talk

2025	T28	Guest lecture for Calculus I (MAT183) , Universiti Teknologi Mara (UiTM) Sarawak Branch Calculus: Differentiation and its Applications (A statistical perspective)
	T27	Seminar talk on R programming , Authority for Info-communications Technology Industry (AITI) R for Data Science
2024	T26	KAUST STAT Workshop 2024 , KAUST, Kingdom of Saudi Arabia Leveraging INLA for estimating Bayesian latent variable models
	T25	Department of Statistics and Data Science Seminar , NUS, Singapore Weighted pairwise likelihood goodness-of-fit tests for binary factor models
	T24	Mathematical Science Departmental Seminar , UBD, Brunei Item Response Theory (IRT) models: Reducing bias in small samples
	T23	International Meeting of the Psychometric Society (IMPS) , Prague University of Economics and Business, Czech Republic Empirical bias-reducing adjustments for Item Response Theory (IRT) models
	T22	Adolphe Quetelet Seminar Series , Ghent University, Belgium Weighted pairwise likelihood and limited information goodness-of-fit tests for binary factor models
2023	T21	Brunei R User Group Meetup , UBD Startup Centre Analysing spatial data using R
	T20	Social Statistics Seminar , LSE, London, UK Spatio-temporal modelling of property prices in Brunei Darussalam
	T19	Mathematical Science Departmental Seminar , UBD, Brunei Pairwise likelihood goodness of fit tests for binary factor models
	T18	International Meeting of the Psychometric Society (IMPS) , University of Maryland, USA Pairwise likelihood goodness of fit tests for binary factor models
	T17	Teaching and Learning Centre Workshop , UBD, Brunei Quality Typography in Teaching Material: An interactive introduction to LaTeX
2022	T16	Department of Statistics and Data Science Seminar , NUS, Singapore Regression modelling using I-priors
	T15	International Meeting of the Psychometric Society (IMPS) , University of Bologna, Italy Selecting interaction effects in additive models using I-priors
	T14	MINDEF Scholars Sharing Session , Online Life After MINDEF
2020	T13	IADA Seminar , UBD, Brunei A latent variable model for maximal performance testing with dropouts for military applications
	T12	Social Statistics Group Meetings , LSE, London, UK Investigating the effect of load carriage on soldiers' performances using structural equation models

2019	T11	UBDSBE Seminar Invitation , UBD, Brunei Bayesian Variable Selection for Linear Models
	T10	CSTRAD Sharing Session , Ministry of Defence, Brunei Misconceptions in Demography
2018	T09	FOS Research Seminar , UBD, Brunei A Brief Guide to Variational Inference
	T08	Inaugural Workshop on Data Science Theory and Practice , LSE, London, UK Binary and Multinomial Regression using Fisher Information Covariance Kernels (I-priors)
	T07	Social Statistics Group Meetings , LSE, London, UK A Beginner's Guide to Variational Inference
2017	T06	Turing Data Study Group , The Alan Turing Institute, London, UK Can data science help identify potential drivers of extremism?
	T05	International Meeting of the Psychometric Society (IMPS) , University of Zürich, Switzerland Regression Modelling with I-Priors
	T04	Department of Statistics PhD Presentation Event , LSE, London, UK Binary probit regression with I-priors
2016	T03	Social Statistics Group Meetings , LSE, London, UK I-priors in Bayesian Variable Selection: From Reproducing Kernel Hilbert Spaces to Hamiltonian Monte Carlo
2015	T02	Postgraduate Research Seminar , LSE, London, UK Two-stage Bayesian variable selection for linear models using I-priors
	T01	Department of Statistics PhD Presentation Event , LSE, London, UK Regression Modelling using I-priors

Skills

🔧 **Programming, data analysis and visualisation** using R, BUGS/JAGS, Stan, Stata, Mplus, and Mathematica

🔬 **Reproducible research** using Quarto, R Markdown, LaTeX, Git and AWS EC2

💻 **Web development** using HTML, CSS, Hugo, GitHub Pages and Netlify

Made with the R package pagedown. The source code is available from github.com/haziqj/cv.
Document last rendered on 2025-11-06 09:27:06.446333 [UTC].

Links

1. <http://stats.lse.ac.uk/bergsma/> ↗
2. <http://stats.lse.ac.uk/moustaki/> ↗
3. <https://phd.haziqj.ml/> ↗

4. [http://www2.warwick.ac.uk/fac/sci/statistics/staff/academic-research/firth/↵](http://www2.warwick.ac.uk/fac/sci/statistics/staff/academic-research/firth/)
5. <https://community.amstat.org/businessandeconomicstatisticssection/new-item/new-item2↵>
6. <https://scholarship.mindef.gov.bn/Theme/Home.aspx↵>
7. <http://bayescomp.kaust.edu.sa/↵>
8. <http://lse.ac.uk/Statistics/↵>
9. <http://fos.ubd.edu.bn/↵>
10. <https://ubdmaths.github.io/dy/↵>
11. <http://lse.ac.uk/Statistics/↵>
12. <https://www2.mindef.gov.bn/cstrad/↵>
13. <https://news.lockheedmartin.com/2011-12-13-Sikorsky-Signs-with-Brunei-Ministry-of-Defence-for-Sale-of-S-70iTM-BLACK-HAWK-Helicopters↵>
14. <https://bruneir.github.io↵>
15. <https://imstat.org/journals-and-publications/annals-of-applied-statistics/↵>
16. <https://rss.onlinelibrary.wiley.com/journal/14679876↵>
17. <https://www.frontiersin.org/journals/applied-mathematics-and-statistics/↵>
18. <http://www.utb.edu.bn/↵>
19. <https://www.instagram.com/ablebrunei/↵>
20. <https://www.r-consortium.org/all-projects/r-user-group-support-program↵>